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Task-01

1) Difference between "Love and Love" in NPL.

→ In NPL, "Love" and "Love" are treated as two different tokens because of case sensitivity. This can lead to duplication of feature and inconsistency in analysis. To avoid this, text is usually converted to lowercase during preprocessing so that both words are treated as the same token.

2) What happen if stopword are not removed?

→ If stopwords are not removed, that dataset may contain a lot of unnecessary and frequently occurring words like "is", "the", and "and". This increases noise, make the model less efficient, and can reduce accuracy because meaningful words get less importance.

3) Two real-world sunarios where removing stopwords can be harmful.

→ Removing stopwords can be harmful in sentiment analysis, where words like "not" change the meaning of a sentence (eg., "not good" vs "good") It can also affect question-answering systems, where words like "what", "who", and "when" are

important for understanding the query.

4) Difference between stemming and lemmatization.

→ Stemming is a simple technique that removes suffixes from words to reduce them to their base form, often without considering context (eg., "running" becomes "run") Lemmatization is a more advanced method that uses vocabulary and grammer rules to convert words into their meaningful root form (eg. "better" becomes "good"). Lemmatization is more accurate but computationally expensive.